Evaluation of Diabetes Treatment Interventions

This randomized controlled trial compared the success rates of three interventions: met, rosi, and lifestyle. The observed success rates were 48.28% for met, 61.37% for rosi, and 53.42% for lifestyle. These initial results suggest that rosi performed best, followed by lifestyle and met. To assess uncertainty in these comparisons, 10,000 bootstrap samples were drawn independently within each treatment arm. Pairwise treatment effects were calculated as differences in success rates, and percentile bootstrap confidence intervals were used to evaluate the evidence for differences between interventions.

The strongest evidence of a difference was between met and rosi. The estimated effect, met minus rosi, had a 95% confidence interval of [−0.221, −0.041], which excludes zero. This indicates that met had a statistically significantly lower success rate than rosi at the 5% significance level. The interval corresponds to an estimated disadvantage for met of approximately 4.1 to 22.1 percentage points. As seen in Table 1, the 90%, 95%, and 99% confidence intervals all exclude zero, so this finding persists even at the more stringent 1% significance level. These results provide evidence against adopting met in preference to rosi based on treatment success alone.

The evidence comparing met with lifestyle was inconclusive. The 95% confidence interval for met minus lifestyle was [−0.142, 0.039], which includes zero. Although the observed success rate was higher for lifestyle than for met, the trial does not establish a statistically significant difference between these interventions at the 5% level. The data are therefore insufficient to conclude that either intervention is more effective than the other. Similarly, the 95% confidence interval for rosi minus lifestyle was [−0.010, 0.165], which also includes zero. Although rosi had a higher observed success rate, the evidence does not establish that it is more effective than lifestyle at the 5% level.

Overall, rosi appears to be the most promising intervention based on its highest observed success rate and its statistically significant advantage over met. However, the trial does not provide conclusive evidence that rosi outperforms lifestyle. The bootstrap KDE and ECDF plots of the pairwise treatment effects (Figure 1) show the distributions of estimated differences across 10,000 bootstrap samples, illustrating the uncertainty in these comparisons relative to zero, which represents no difference in success rates. The met − rosi distribution is concentrated below zero, consistent with the confidence interval evidence, whereas the other two comparisons have distributions that span zero. Therefore, while rosi is a promising candidate, the evidence is insufficient to conclude that it is more effective than lifestyle. Decisions about adopting an intervention should also consider treatment costs, adverse effects, patient preferences, and clinical outcomes beyond the binary success measure. These conclusions are based on individual pairwise confidence intervals, without adjustment for multiple comparisons.

Pairwise comparison	90% CI	95% CI	99% CI
Met − Rosi	[−0.208, −0.058]	[−0.221, −0.041]	[−0.247, −0.011]
Met − Lifestyle	[−0.129, 0.026]	[−0.142, 0.039]	[−0.172, 0.064]
Rosi − Lifestyle	[0.003, 0.152]	[−0.010, 0.165]	[−0.036, 0.191]

Table 1: Percentile bootstrap confidence intervals for pairwise treatment effects

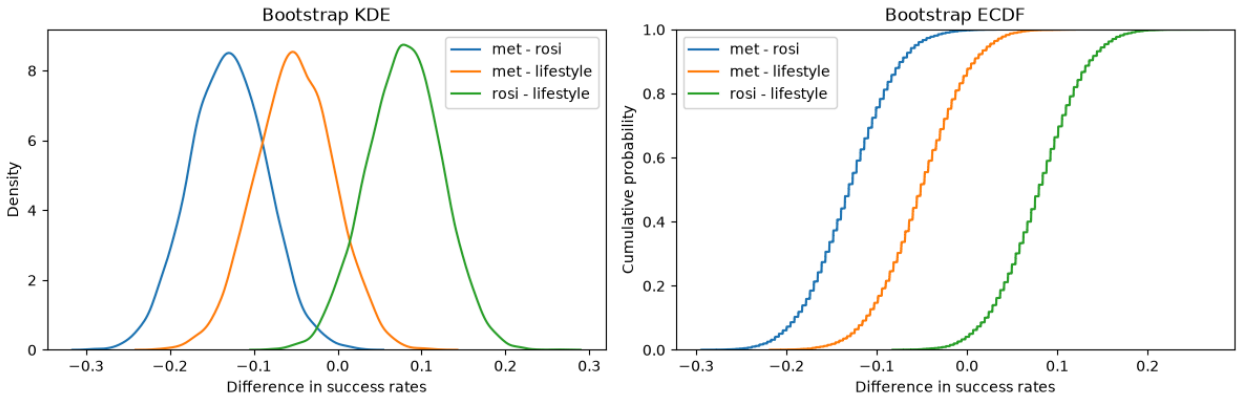


Figure 1: Bootstrap KDE and ECDF plots of pairwise treatment effects.